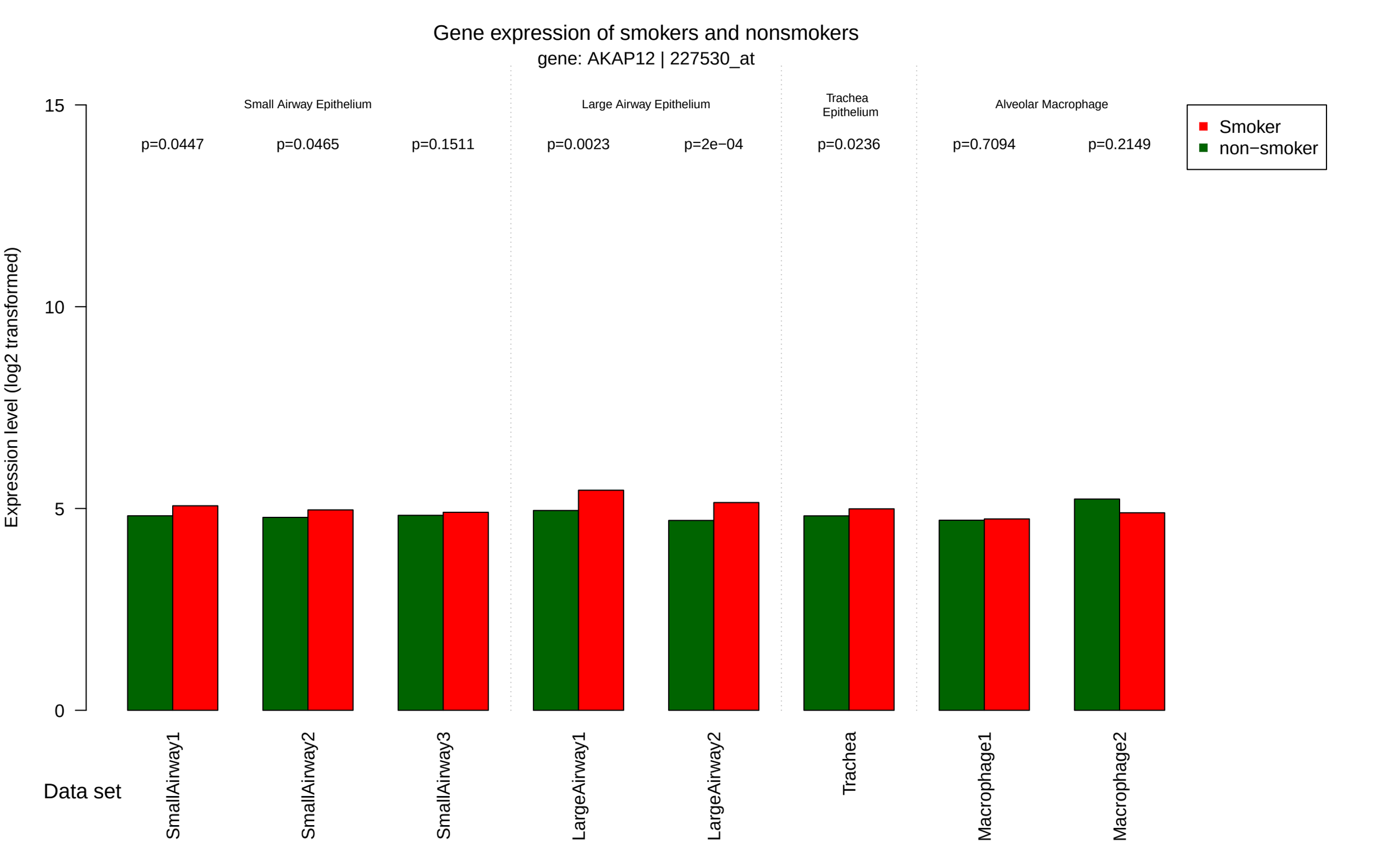
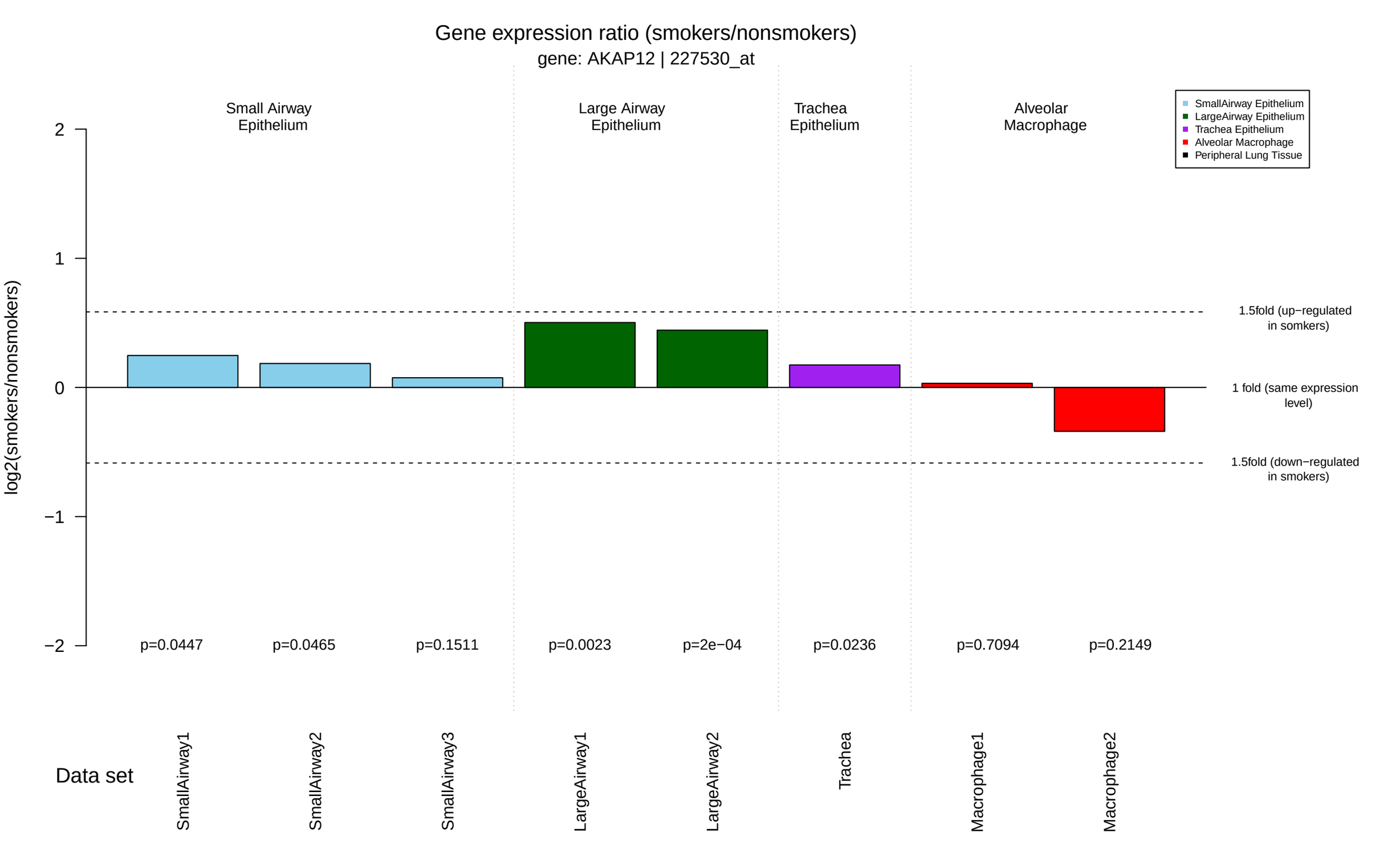
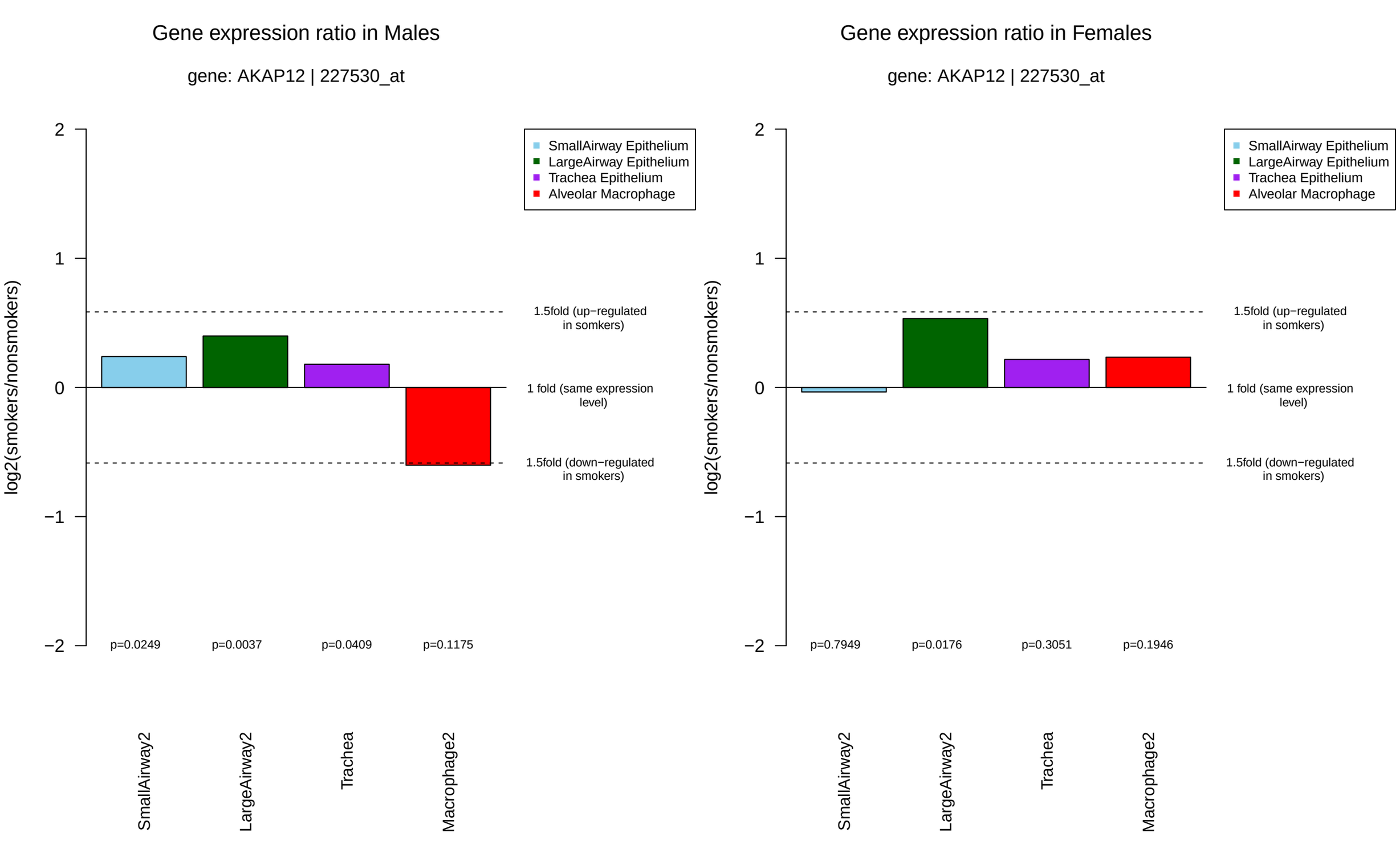
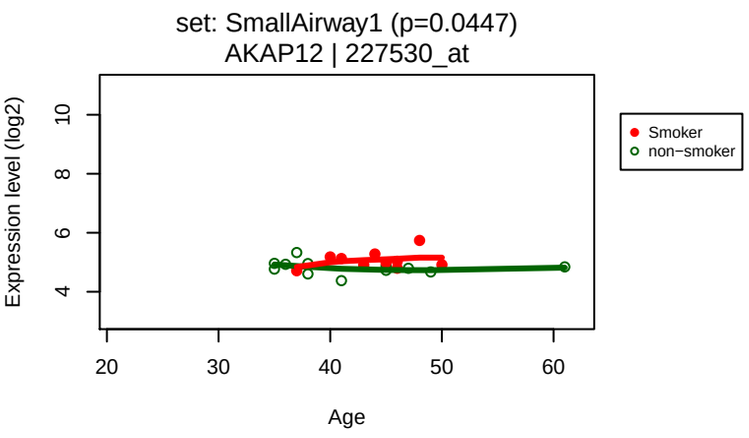
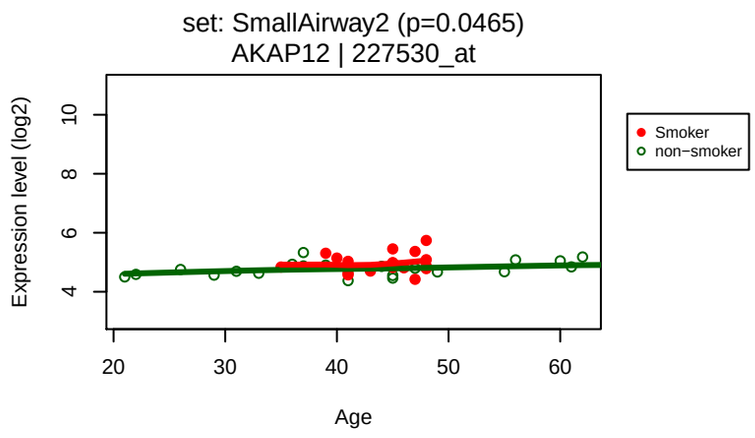
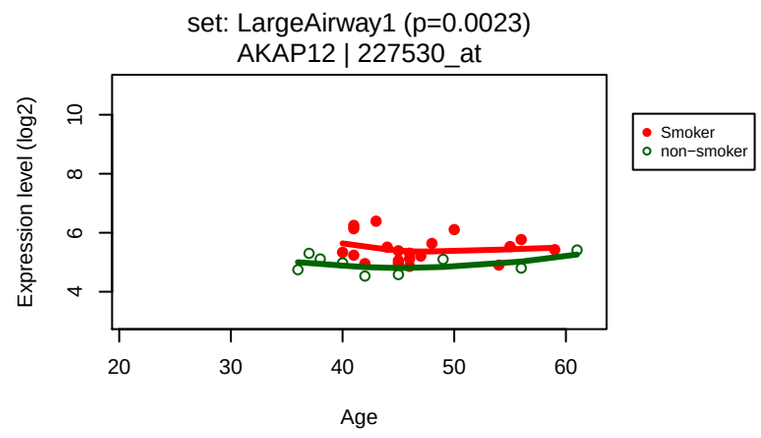

Gene expression of smokers and nonsmokers
gene: AKAP12 | 227530_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0447
p=0.0465
p=0.1511
p=0.0023
p=2e−04
p=0.0236
p=0.7094
p=0.2149
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

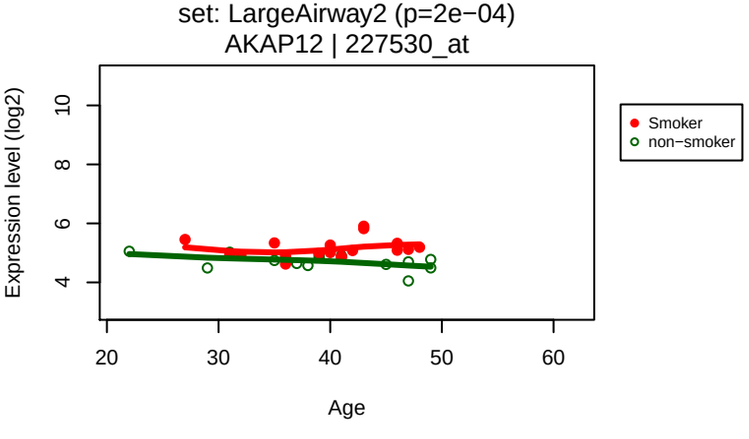

Gene expression ratio (smokers/nonsmokers)
gene: AKAP12 | 227530_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0447
p=0.0465
p=0.1511
p=0.0023
p=2e−04
p=0.0236
p=0.7094
p=0.2149
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

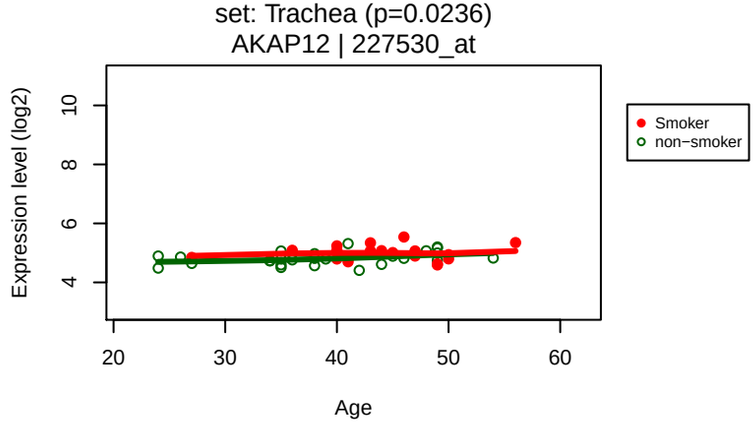

Gene expression ratio in Males
gene: AKAP12 | 227530_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0249
p=0.0037
p=0.0409
p=0.1175
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKAP12 | 227530_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7949
p=0.0176
p=0.3051
p=0.1946
SmallAirway2
LargeAirway2
Trachea
Macrophage2

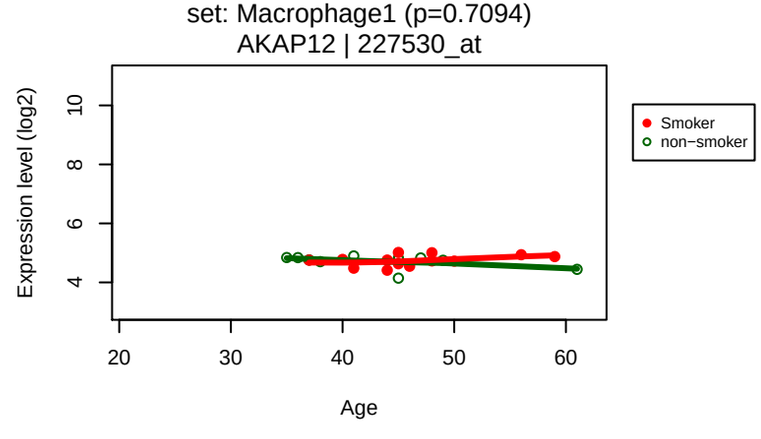

set: SmallAirway1 (p=0.0447)
AKAP12 | 227530_at
Expression level (log2)
Age
Smoker
non−smoker

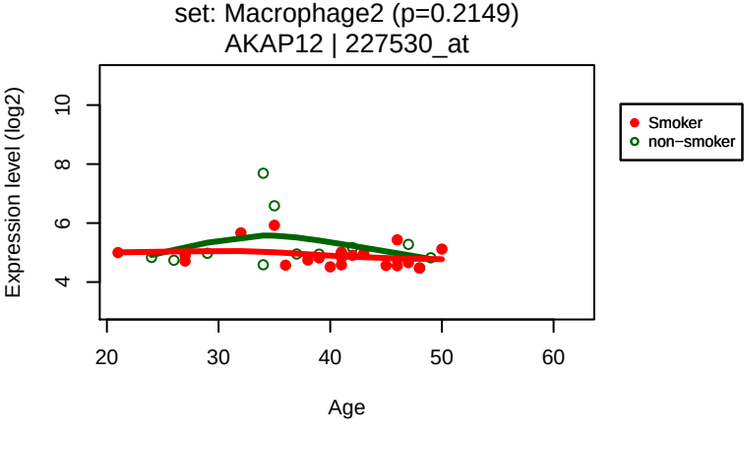

set: SmallAirway2 (p=0.0465)
AKAP12 | 227530_at
Expression level (log2)
Age
Smoker
non−smoker

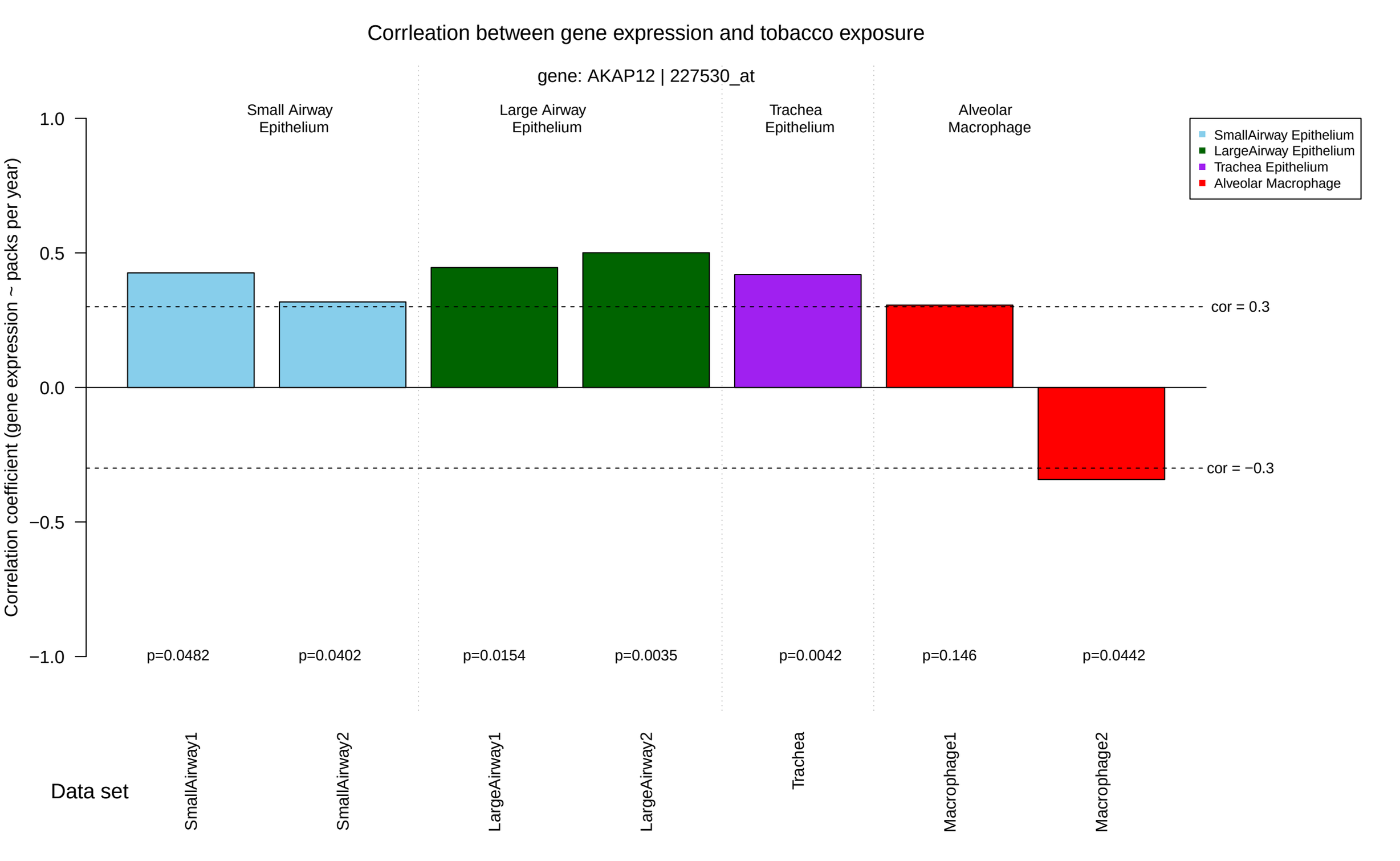

set: LargeAirway1 (p=0.0023)
AKAP12 | 227530_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=2e−04)
AKAP12 | 227530_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0236)
AKAP12 | 227530_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7094)
AKAP12 | 227530_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2149)
AKAP12 | 227530_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKAP12 | 227530_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0482
p=0.0402
p=0.0154
p=0.0035
p=0.0042
p=0.146
p=0.0442
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2